DEM super-resolution assisted by remote sensing images content feature

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Abstract: **Deep learning has achieved promising progress in the super-resolution (SR) of digital elevation models (DEMs) by effectively exploiting the correspondences of terrain features at different spatial scales. However, the existing deep learning based DEM SR methods rarely consider the integration of the available multi-modal data, which can provide auxiliary high-frequency information for resolution enhancement. In this paper, we explore the integration of optical remote sensing images for assisting DEM SR, where the images have a higher resolution than the DEM dataset to be reconstructed. This is achieved by constructing a terrain-guided content feature learning network, where the core idea is to transfer the feature representation of high-resolution image textures with the guidance of informative terrain features, for adapting DEM SR learning. Specifically, a content feature learning module is established to calibrate the content details to the learned terrain feature space, by exploiting the multi-dimensional attention mechanism. Extensive experiments were conducted on simulated datasets. The results indicated that the proposed method can reconstruct a DEM with high-fidelity elevation and sharper terrain details by incorporating the complementary information from satellite images, and outperforms the competitive comparison methods.**

Keywords: DEM super-resolution; data fusion; remote sensing imagery; deep learning.

I. Introduction

High-resolution (HR) digital elevation models (DEMs) can provide fine-scale topographic information in a digital representation, which is of great significance for accurate modeling and analysis in geoscience field [1]. As remote sensing techniques have advanced rapidly, satellite observations have facilitated the generation of diverse global-scale DEMs. However, the spatial resolution of the freely available DEM datasets is limited to 1 arc-second (~30 m) over the globe, which limits their potential in applications requiring fine-scale terrain information [2].

To tackle the issue of the resolution limit, the simplest solution is to use spatial interpolation to process the low-resolution (LR) DEM. However, the interpolation methods cannot reconstruct the real high-frequency details in the results without supplementary information, and result in an oversmoothed terrain surface over rough areas [3]. Super-resolution (SR) provides a cost-effective technique to reconstruct an HR DEM with finer terrain features by blending information from LR DEM. As distinguished from the interpolation methods, the essence of SR is to integrate the information from multi-source data via fusion techniques. With the explosive development of deep learning techniques, researchers have explored the use of deep learning models for the SR task of DEM [4-6]. However, most of methods only consider the complementary information of homogeneous data to improve the resolution, vertical accuracy and spatial continuity of data, which is difficult to fundamentally solve the quality problem of altimetry data affected by multiple factors such as observation technology, generation method and terrain conditions.

In this paper, we investigate the exploitation of remote sensing images for assisting DEM reconstruction, and propose a novel SR framework based on a content feature learning model. The key issue is to fully utilize the high-frequency information of the higher-resolution remote sensing images with the guidance of terrain features for enhancing the spatial resolution of the DEM, by exploring their discriminative feature representations. The model's inputs include the LR DEM to be reconstructed and the HR image as auxiliary data. Specifically, the structural features extracted from the DEM and the image are fused via a residual spatial-channel attention block (RSCAB) to form the topographic textures within content feature learning module. Finally, the terrain-oriented structures and LR DEM are skip connected to generate the SR result. A number of experiments were carried out on simulated DEM datasets, where the classic interpolation methods and the state-of-the-art deep learning-based SR methods were used for comparison.

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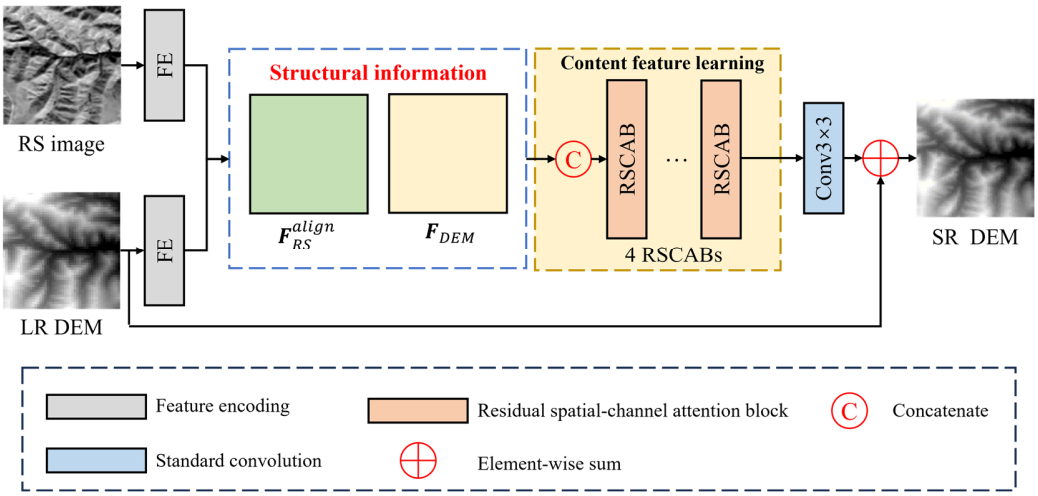

Figure 1 Overview of the flowchart of the proposed method.

II. Method

The overall flowchart of the proposed method is presented in Figure. 1. Firstly, the remote sensing image and DEM are processed by a feature encoding module consisting of four 3×3 standard convolutional layers to obtain image features (F_{RS}) and terrain features (F_{DEM}). In this way, the heterogeneous data recorded as spectral reflectance and terrain elevation in the raster grids can be transformed into feature maps represented in the hidden layer space.

A. Model structure

In the content feature learning module, the structural information from the input DEM and image is fused to form the base texture of the SR result. Inspired by the work of Woo et al. and Zhang et al. [7], [8], we propose the residual spatial-channel attention (RSCA) mechanism to retain the visual details delineating regional topographic relief. The combination of residual learning and spatial and channel attention in the proposed module can serve to accurately select the informative components that commonly exist in the feature channels.

The structure of the RSCAB is shown in Figure. 2. Given the concatenated features $F = [F_{DEM}, F_{RS}]$ as the input, the residuals are obtained via residual learning, which can be formulated as:

$$R_n = f_{3\times3}(\delta(f_{3\times3}(F_{n-1}))) \tag{1}$$

where F_{n-1} and R_n are the input and residual output of the nth RSCAB, respectively. Moreover, $f_{3\times3}$ denotes a 3×3 convolutional layer, and $\delta(\cdot)$ is the leaky rectified linear unit (ReLU) function. The residuals are then rescaled with the spatial-channel attention process, which can be expressed as:

$$R_s = R_n \otimes M_s,\ R_c = R_s \otimes M_c \tag{2}$$

where M_s and M_c denote the spatial and channel attention maps, respectively. $\otimes$ indicates the element-wise multiplication. In this study, the basic assumption is that perceptual topographic structural features can be observed in both the imagery and DEM. Therefore, larger weights are assigned to the pixels, based on spatial attention weights (M_s). Moreover, the terrain structural features extracted from the visible imagery with a higher resolution provide sharper details that should contribute more to the fusion results, and the channel attention map (M_c) serves for channel-wise feature rescaling. Specifically, the M_s is computed as:

$$M_s = \sigma(Avgpool_c(R_n)) \tag{3}$$

where $Avgpool_c(\cdot)$ is the average pooling operation along the channel dimension and $Avgpool_c(R_n)$ obtains a two-dimensional feature map, while $\sigma(\cdot)$ is the sigmoid activation function. The channel attention map can be computed as:

$$\begin{aligned} M_c &= \sigma(f_{1\times1}\left(\delta\left(f_{1\times1}(Avgpool_s(R_s))\right)\right) \\ &+ f_{1\times1}\left(\delta\left(f_{1\times1}(Maxpool_s(R_s))\right)\right)) \end{aligned} \tag{4}$$

where $Avgpool_s(\cdot)$ and $Maxpool_s(\cdot)$ indicate the average pooling and max pooling operations. $f_{1\times1}(\cdot)$ is a 1×1 convolutional layer. To better learn the residual information through the network, a short skip connection is used to get the output of the RSCAB via:

$$F_n = F_{n-1} + R_c \tag{5}$$

There are four RSCABs in the content feature learning module. The output of the content feature learning module is denoted as $F_{texture}$. Finally, the residual component obtained by the content feature learning module is added back to the original LR DEM, which can be expressed as:

$$I_{SR} = F_{fusion} + I_{DEM} \tag{6}$$

where I_{SR} is the HR DEM output obtained by the proposed method.

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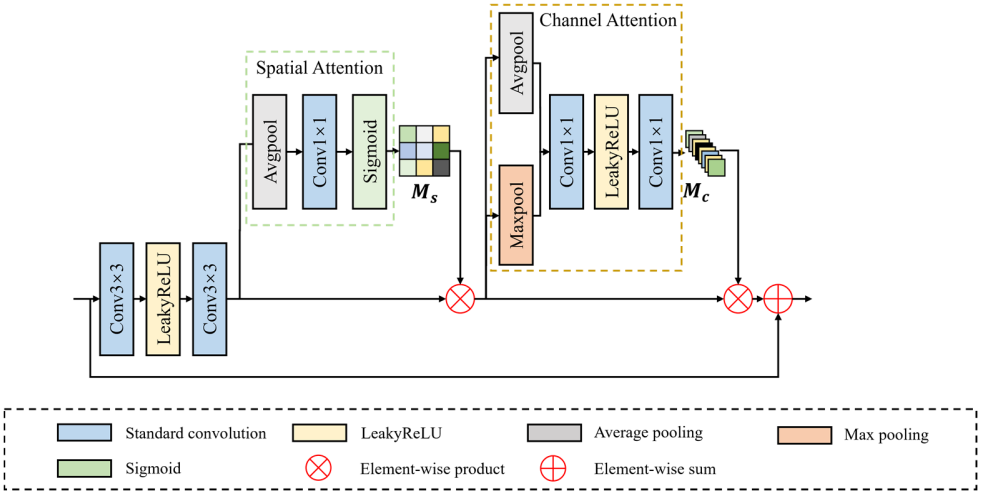

Figure 2 The architecture of the residual spatial-channel attention block (RSCAB)

B. Loss function

The loss function of our network consists of two parts, namely global content loss and edge loss. All losses are calculated based on the Charbonnier loss, which can handle outliers better and improve SR performance over L_2 loss functions.

The global content loss computes the average statistics of the pixel-wise differences between the SR result with the reference HR DEM, which promotes the global accuracy of the reconstructed DEM. The formula for the global content loss can be expressed as follows:

$$L_{SR} = (\|I_{SR} - I_{HR}\| + \varepsilon^2)^{1/2} \tag{7}$$

where I_{SR} and I_{HR} are SR and HR images respectively, and ε is a small constant value, which is set to 1×10^{-6} in our experiments.

Given that high-frequency terrain features play the dominant role in determining the accuracy of DEM analysis, we defined a loss term with the combined minimization of the differences between the terrain features of the super-resolved DEM and the reference. The calculation formula of edge loss is shown as follows:

$$L_{Edge} = (\|\partial(I_{SR})\text{-}\partial(I_{HR})\| + \varepsilon^2)^{1/2} \tag{8}$$

where $\partial(\cdot)$ represents slope extraction operator. The formula for calculating slope is as follows:

$$dx = \left(z_7 - z_1 + \sqrt{2}(z_8 - z_2) + z_9 - z_3\right) / \left(\left(4 + 2\sqrt{2}\right) \times Cellsize\right) \tag{9}$$

$$dy = (z_3 - z_1 + \sqrt{2}(z_6 - z_4) + z_9 - z_7) / \left(\left(4 + 2\sqrt{2}\right) \times Cellsize\right) \tag{10}$$

$$slope_{ij} = arctan(dx^2 + dy^2)^{1/2} \tag{11}$$

where $e_{i,j}$ represents the elevation value at $Cell(i,j)$, $Cellsize$ represents the resolution of DEM, $slope_{ij}$ represents the slope at $Cell(i,j)$.

The formula for calculating the total loss is as follows:

$$L_{Total} = L_{SR} + \alpha L_{Edge} \tag{12}$$

where α is a hyper-parameter, which is set as 1 in the experiment.

III. Experiment

A. Dataset

In this study, we collected five TanDEM-X data scenes with a 12-m resolution covering different land covers and terrain conditions, where the size of each was 9001×9001 pixels. The Sentinel-2 images were collected as the auxiliary modality for DEM reconstruction. To match the TanDEM-X DEM data, cloudless Sentinel-2 images with minimum terrain shadows acquired between 2014–2015 were selected accordingly. Given that the images play the role of providing the detailed ground textures, the red band with a 10-m spatial resolution and high signal-to-noise ratio (SNR) was used in this study. In order to improve training efficiency, we cut the data into 22750 144×144 data patches, in which the ratio of training set to test set is about 8:2.

The code for our method is implemented using the Pytorch framework with an NVIDIA 3090 GPU. The initial learning rate is 2×10^{-4} and halved at 50 epochs. The model is trained for 100 epochs, the batchsize is 32. We compared the proposed method with three methods (i.e., Bicubic, SRCNN [9] and TfaSR [10]).

B. Comparison to the State-of-the-arts method

When applying the proposed method and comparing it with the aforementioned several comparative methods on the test data. The evaluation metric results, as shown in Table 1, indicate that the proposed method outperforms the other methods in each evaluation metrics. Bicubic performs much worse than other deep learning methods, which also proves the effectiveness of deep learning methods.

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Table 1 Quantitative comparisons of the State-of-the-art methods.

Model	Scale	RMSE	MAE	RMSE-Slope
Ours	×3	1.031	0.660	2.608
TfaSR		1.141	0.769	2.903
SRCNN		1.314	0.961	3.031
Bicubic		1.404	0.960	4.024

It can be seen from Figure 3 that in some areas with complex terrain, the proposed method can distinguish the edge features, and the terrain trend of the reconstruction results is consistent with the real surface. The experimental results indicate that, compared to other methods, this approach preserves more terrain details within the test area and recovers the true distribution pattern of the DEM.

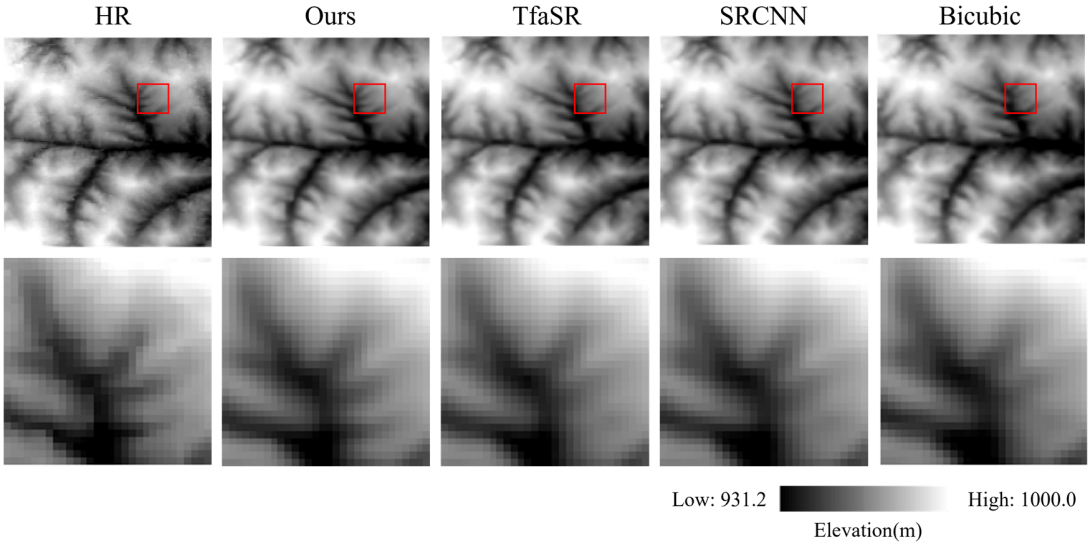

Figure 3 The comparison of different methods

C. Ablation study

In this section, we replace the remote sensing images in the network input with DEM for experiments to prove the help of remote sensing images to the network performance. As can be seen from the results in the Table 2, after changing the input data from remote sensing images to LR DEM, the network performance also shows a certain decrease. The results show that each part of the network plays a role in improving the network performance.

Table 2 Quantitative comparisons by the proposed method with different settings for 3×SR.

Input	RMSE	MAE	RMSE-Slope
DEM+RS	1.031	0.660	2.608
DEM+DEM	1.083	0.702	2.756

IV. Conclusion

In this paper, we have proposed a content feature learning network for DEM SR, with the aid of auxiliary remote sensing imagery. As distinguished from most of the existing SR frameworks, we introduce the high-frequency information from an HR remote sensing image with the target resolution. To integrate the multi-modal information within the DEM and image, the textures are extracted from the feature maps of the DEM and the image in the content feature learning module.

The simulated experiments on TanDEM-X datasets showed that the proposed method can achieve superior evaluation metric scores and visual results with richer fine-scale details than the representative CNN-based SR methods and the traditional interpolation method. The research results can effectively improve the spatial resolution and the continuity of terrain features of DEM data products, and make them better serve terrain and geomorphology analysis, land surface process modeling and industrial applications related to geographic information.

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